

Technologické aplikace bezpečnosti

Hacking the Point Cloud: LiDAR Security in Autonomous Vehicles

Šimon Semerád



České vysoké učení technické v Praze,
Fakulta informačních technologií
Katedra informační bezpečnosti

Introduction to LiDAR

- **LiDAR** (Light Detection and Ranging) is a primary spatial mapping sensor in autonomous vehicles (AVs)
- Functions by emitting hundreds of thousands of laser pulses (typically infrared) per second and capturing their reflections
- **Time of Flight (ToF)**: The core principle relies on measuring the exact time t a photon takes to travel to an object and back
- The distance d is calculated using the constant speed of light c :

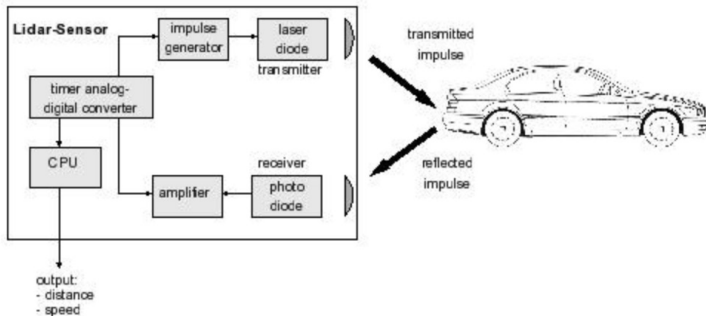
$$d = \frac{c \cdot t}{2}$$

- A typical 3D LiDAR system consists of distinct hardware components:
 - ▶ **Transmitter:** An impulse generator coupled with a laser diode emits the signal
 - ▶ **Receiver:** A highly sensitive photodiode captures the reflected photons
 - ▶ **Signal Processing:** The weak returned signal passes through an amplifier and an analog-to-digital converter (ADC).
 - ▶ **CPU:** Computes the final distance and extracts relative speed (via distance change over time) and signal intensity

LiDAR Architecture – Diagram

Functional Structure of a Lidar-Sensor

ika



12H0034.ppt

33N0147.ppt

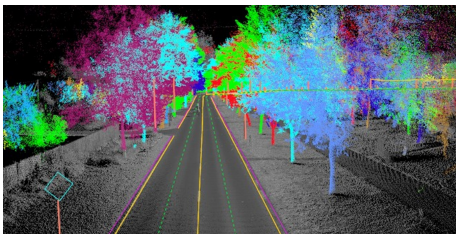
Source: <https://www.microcontrollertips.com>

Output Data & HD Mapping

- The CPU processes the ToF data to generate a 3D Point Cloud (precise X, Y, Z spatial coordinates)
- **Intensity (Reflectivity):** The sensor also measures the strength of the returned signal
- **Retroreflective objects**, such as traffic signs and lane markings, yield significantly higher intensity values
- **Localization:** Since standard GPS lacks centimeter-level precision, AVs match these high-intensity reflections against pre-recorded High-Definition (HD) Maps to pinpoint their exact location

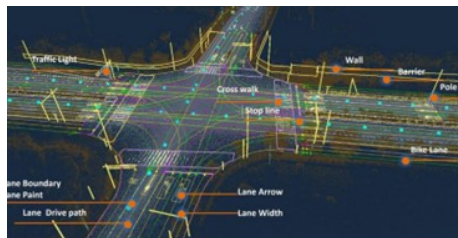
Output Data & HD Mapping – Visualized

LiDAR Point Cloud



Source: encrypted-tbn0.gstatic.com

High-Definition (HD) Map



Source: cdn.open-pr.com

Hardware-Level Vulnerability

- Despite advanced data processing, the LiDAR unit remains fundamentally a "blind" physical sensor.
- **The Core Flaw:** The system inherently trusts that any received photon is a legitimate reflection of its own emitted pulse.
- Historically, it lacks the ability to authenticate incoming light signals.
 - ▶ *Note:* While modern countermeasures (e.g., pulse randomization, optical watermarking) are emerging, many deployed systems remain vulnerable.
- This physical-layer trust makes the sensor highly susceptible to cyber-physical attacks:
 - ▶ **Spoofing:** Injecting false signals to create non-existent obstacles.
 - ▶ **Shadowing:** Suppressing or masking legitimate reflections.

- Item 1
- Item 2
- Item 3

Figure and Columns

- Item 1
- Item 2
- Item 3



Zdroj: <http://example.com>

Listings

```
//Before retpoline
jmp *%rax

//After retpoline
1: call load_label
capture_ret_spec:
2: pause; lfence
3: jmp capture_ret_spec
load_label:
4: mov %rax, (%rsp)
5: ret
```